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Studierichting: Informatica
Oefeningenreeks 6E nr. 1.2

$$T_7(f, 0)(x) \text{ voor } f(x) = \tan x$$

1. $f(x) = \tan x$
2. $f'(x) = \sec^2 x \Rightarrow f'(0) = 1$
3. $f''(x) = 2 \sec^2 x \tan x \Rightarrow f''(0) = 0$
4. $f^{(3)}(x) = 2 \sec^2 x (\sec^2 x + 2 \tan^2 x) \Rightarrow f^{(3)}(0) = 2$
5. $f^{(4)}(0) = 0$
6. $f^{(5)}(0) = 16$
7. $f^{(6)}(0) = 0$
8. $f^{(7)}(0) = 272$

$$\begin{aligned} T_7(f, 0)(x) &= f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \dots + \frac{f^{(7)}(0)}{7!}x^7 \\ &= 0 + \frac{1}{1}x + \frac{0}{2}x^2 + \frac{2}{6}x^3 + \frac{0}{24}x^4 + \frac{16}{120}x^5 + \frac{0}{720}x^6 + \frac{272}{5040}x^7 \\ &= x + \frac{1}{3}x^3 + \frac{2}{15}x^5 + \frac{17}{315}x^7 \end{aligned}$$

References